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Project Title:	High LET Radiation And Neurogenesis: Implications And Mechanisms Underlying Cognitive Impairment		
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Zip Code:	92697-2695	**Congressional District:**	45
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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	For those engaged in space travel, the radiation environment poses a range of potential health risks. For instance, it is possible that exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can induce a range of CNS injuries, including cognitive impairment. While the pathogenesis of radiation-induced cognitive impairment is not clear, we have shown that neuronal precursor cells in the hippocampus may be involved. Using low doses of X-rays (1-5 Gy), we have demonstrated, in vitro and in vivo, that neural stem cells of the hippocampus are exquisitely sensitive to irradiation. In addition, these cells exhibit oxidative stress and undergo apoptosis after irradiation. Low dose x-irradiation of these cells in vitro also leads to a persistent oxidative stress that coincides temporally with a dose-dependent reduction in neurogenesis observed in vivo. Consequently, we hypothesize that oxidative stress plays a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo. Further, we suggest that oxidative stress constitutes a biochemical mechanism that can influence the fate of stem cell differentiation		

Task Description:	in vitro and hippocampal neurogenesis in vivo. It is our overall objective to determine how oxidative stress affects the high LET radiation response of precursor cells in the subgranular zone (SGZ) of the hippocampus and to identify compounds/strategies that will enable us to ameliorate the adverse effects of high LET irradiation on hippocampal neurogenesis. We have defined 4 specific aims to address our objective. Aim 1: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to acute (days) oxidative stress and altered neurogenesis after exposure to high-energy protons. Aim 2: Using in vitro and in vivo models, determine if deficiencies in site-specific superoxide dismutases (SODs) lead to chronic (months) oxidative stress and altered differentiation/neurogenesis following exposure to high-energy iron ions. Aim 3: Determine if antioxidant treatment with SOD/catalase mimetic agents reduce high-LET induced radiation injury to neural stem cells in vitro and ameliorate the radiation-induced inhibition of neurogenesis in vivo. Aim 4: Quantify the effects of high-energy iron ions on cognitive function in wild type mice and determine if antioxidant treatment ameliorates those effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Elucidating oxidative stress as a biochemical mechanism regulating the behavior of neural precursor cells will provide significant insight into how irradiation might lead to impaired cognition during spaceflight. These studies will also provide information relevant to the development of antioxidant strategies to ameliorate the adverse effects of irradiation in space.
Task Progress:	Summary of Accomplishments: Progress over the four years of funding has been significant, as evidenced by 8 published papers. NASA funding has laid the groundwork for a number of studies that have now clearly established a significant role for oxidative stress in modulating the CNS response to high and low LET ionizing radiations. These studies were the first to demonstrate the space relevant radiation doses (= 1 Gy) led to significant increases (>10-fold) in oxidative stress that persisted (> 2 months) for long-periods after initial exposure. Corroborating in vivo work has established a temporal correlation between in vitro endpoints, and has helped to validate the importance of oxidative injury and secondary reactive processes in the brain that contribute to impaired neurogenesis and cognition. NASA funding has been critical in advancing our knowledge of the response of the CNS to heavy ion irradiation. Our proof of principle studies have now demonstrated the importance of oxidative processes in the irradiated CNS, and provide the necessary framework for advancing these studies further, to elucidate the mechanistic basis of cognitive impairment in the CNS that will be critical in establishing valid risk estimates. To date we have undertaken 6 runs at BNL (two runs in 2005, 2006 and 2007) as well as two runs at Loma Linda. We have now successfully irradiated a large number of in vitro samples as well as our SOD3 knockout mice, mice heterozygous for all three SOD isoforms (CuZnSOD, MnSOD and SOD3) and all relevant WT control mice. With these studies, we continue to make significant progress in completing the objectives outlined in this grant. Major accomplishments are summarized below: 1) In vitro studies using hippocampal and subventricular neural precursor cells have revealed that 56Fe-ions induce marked levels of oxidative stress. Reactive oxygen species detected after heavy ion irradiation are 2-10 fold higher at various times (6-48h) after irradiation. 2) In vitro studies using hippocampal neural precursor cells have shown protons to be more efficient at eliciting oxidative stress than X-rays. 3) The antioxidant lipoic acid has shown promise at reducing radiation-induced ROS, whether added as a radioprotector (before irradiation) or as a mitigating agent (after irradiation). 4) Comprehensive analysis of our EC-SOD (extracellular SOD) deficient (nullizygous) mice has now shown that these animals are actually protected from the inhibition of neurogenesis after irradiation. While this result is qualitatively similar between low and high LET radiations, heavy ions do show important differences, possibly due to changing levels of activated microglia. This provocative data now shows for the first time that altering the redox status of the microenvironment can be protective to certain adverse radiation effects in the animal. 5) New evidence now reveals that the neuroprotective effects of EC-SOD deficiency are also found in animals deficient for the other two isoforms of SOD. Analysis of Cu/Zn-SOD (the cytosolic isoform) and Mn-SOD (the mitochondrial isoform) deficient mice have show that neurogenesis is also spared after irradiation in these animals. Quantification of immunohistochemical staining continues, but the trend has now been verified statistically, and supports our overall contention that changes in oxidative stress play pivotal roles in the radioresponse of the CNS. 6) Chronic studies analyzing oxidative stress in neural precursor cells that have been exposed to a range of Fe ion doses (0.1 – 2 Gy) have nearly been repeated. This new data has revealed significant increases in oxidative stress that persist over background levels for over two months after initial irradiation. Use of two different fluorogenic dyes now suggests that a fraction of the net increase in oxidative species emanates from the mitochondria. 7) Studies exposing neural precursor cells to Fe ions at very low doses (1-15 cGy) have shown oxidative stress to be significantly elevated over acute times (12-24h) after irradiation. 8) We have recently expanded our low dose studies down to particle fluences spanning the range of 500 - 30,000 Fe particles/cm2. At these ultra-low particle fluences not every cell (or neurosphere) is traversed by an energetic ion particle. However, oxidative stress is still elevated significantly over background. These data suggest that bystander effects are occurring in the irradiated neural precursor cells. Furthermore, the mechanism/s underlying these bystander effects is/are likely to involve oxidative species derived from the small subpopulation of traversed cells, that then impact the entire population to elicit significant increase in oxidative stress.

	9) We have also just recently undertaken a series of studies to determine whether our irradiated neural precursor cells show signs of an adaptive response. In summary, fractionated dosing paradigms result in sparing of oxidative stress when compared to the same dose delivered acutely. Evidence strongly suggests that both reactive oxygen and nitrogen species are involved, and that prior exposure desensitizes these cells to a subsequent dose of heavy ions. 10) Funding from this Grant has resulted in 18 formal presentations at national and international meetings, as well as an assortment of research and teaching seminars at various universities and national laboratories across the country. 11) Funding from this Grant has resulted in 8 published manuscripts and several in preparation.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Limoli CL, Giedzinski E, Baure J, Rola R, Fike JR. "Altered growth and radiosensitivity in neural precursor cells subjected to oxidative stress." Int J Radiat Biol. 2006 Sep;82(9):640-7. PMID: 17050476 , Sep-2006
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Articles in Peer-reviewed Journals	Baulch JE, Craver BM, Tran KK, Yu L, Chmielewski N, Allen BD, Limoli CL. "Persistent oxidative stress in human neural stem cells exposed to low fluences of charged particles" Redox Biol. 2015 Mar 11;5:24-32. [Epub ahead of print] http://dx.doi.org/10.1016/j.redox.2015.03.001 ; PubMed PMID: 25800120; PubMed Central PMCID: PMC4371546 , Mar-2015